

Emotion Analysis Using BRIGHTER + EthioEmo Datasets

Evaluating Data Augmentation Techniques for Multilabel Emotion Classification in Low-Resource African Languages

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1 Introduction

Emotion analysis (EA) automatically identifies textual emotions, but African languages remain underrepresented due to linguistic diversity and data scarcity. Consequently, models struggle with multilabel classification, where a single text conveys multiple overlapping emotions. To mitigate this, this project investigates data augmentation, specifically back-translation and paraphrasing, to artificially expand training datasets. By evaluating these techniques alongside qualitative error analysis, this research seeks to improve multilabel emotion classification for low-resource African languages and build more inclusive, responsible NLP systems.

2 Background

Emotion analysis (EA) identifies specific, often overlapping emotional states (e.g., anger, joy) rather than basic positive or negative feelings, requiring multilabel classification for real-world accuracy (del Arco et al., 2024). However, the vast annotated data required by modern EA models creates a barrier for African languages, which remain heavily low-resource (Adebara and Abdul-Mageed, 2022). While recent Afrocentric efforts have advanced foundational NLP, complex tasks like emotion classification remain under-explored and rarely optimised for typological nuances like complex morphology and code-switching (Adebara, 2024).

To address data scarcity, researchers use data augmentation, specifically back-translation and paraphrasing, to artificially expand training sets. These approaches excel in low-resource environments by increasing lexical diversity and maintaining semantic fidelity, often outperforming AI models without prior examples (Radlinski et al., 2025). However, their application to multilabel emotion classification in African languages remains unexplored. Despite benchmarks like the EthioEmo dataset, the field is still vastly under-resourced.

This project bridges this gap by evaluating back-translation and paraphrasing on multilabel emotion tasks for underrepresented African languages, providing essential insights to build inclusive NLP models.

3 Research Questions

1. How can data augmentation strategies (such as back-translation and paraphrasing) improve multilabel emotion classification for African languages?
2. How can existing NLP methods be adapted or applied effectively in low-resource African language contexts for emotion analysis?

4 Dataset and Data Description

4.1 Primary Dataset: BRIGHTER

This project uses the BRIGHTER dataset, a multilingual collection of approximately 100,000 emotion-annotated text instances across 28 languages. Each instance is labelled with six emotions (joy, sadness, anger, fear, surprise, disgust) and a neutral class, using a four-level intensity scale. The dataset supports multilabel classification, which is the focus of this study.

4.2 Selected Languages

Three African languages are selected: Afrikaans, Hausa, and Igbo. These were chosen based on data availability, societal relevance, and team familiarity. All have complete train, development, and test splits, enabling augmentation experiments. The languages represent distinct language families and together cover a large speaker base, supporting broader applicability of results.

4.3 Confirmed Availability

The dataset is publicly available on Hugging Face under a CC-BY 4.0 licence, and no additional data collection is required.

5 Approach and Methodology

5.1 Overview

This project investigates whether data augmentation improves multilabel emotion classification for low-resource African languages. The methodology consists of baseline model training, data augmentation, augmented model training, and qualitative error analysis.

5.2 Baseline Models

Two pretrained multilingual models are fine-tuned: XLM-RoBERTa-large (general multilingual baseline) and AfroXLMR-large (adapted for African languages). Models are trained separately per language using sigmoid outputs and binary cross-entropy loss for multilabel classification.

5.3 Data Augmentation Techniques

Two augmentation strategies are applied:

1. **Back-translation:** translating text to a pivot language and back to introduce lexical diversity.
2. **Paraphrasing:** generating structurally varied versions of text while preserving meaning.

Training data is expanded to approximately 1.5× and 2× its original size.

5.4 Experimental Conditions

Four conditions are evaluated:

1. Baseline (no augmentation)
2. Back-translation
3. Paraphrasing
4. Combined back-translation and paraphrasing

5.5 Error Analysis

Qualitative error analysis is conducted on the best-performing models. Misclassifications are examined to identify patterns such as emotion overlap, linguistic complexity, and class imbalance, with interpretation informed by language-specific knowledge.

6 Experiments and Results

6.1 Experimental Setup

All experiments are conducted across three language subsets of the BRIGHTER dataset: **Afrikaans**, **Hausa**, and **Igbo**. For Afrikaans, the dataset contains 1,222 training instances, 196 development instances, and 2,130 test instances. For Hausa, the splits consist of 2,145 original training instances (expanded to 2,945 rows post-deduplication handling), 712 development instances, and 2,160 test instances[cite: 1, 2]. For Igbo, the original splits provide 2,880 training instances, 958 development instances, and 2,888 test instances; 800 instances were transferred from the test split to training to mitigate the small training set size, yielding a final training pool of 3,680 instances. Each instance is annotated with up to six binary emotion labels: anger, disgust, fear, joy, sadness, and surprise[cite: brighter dataset here]. For Igbo, the surprise label was excluded from the label space due to near-zero prevalence (2.6% of training instances), and a neutral class was derived implicitly for instances carrying no emotion label.

The training data profiles across these languages exhibit pronounced class imbalance. In the Hausa training partition, for example, sadness is the most dominant emotion (mean label rate ≈ 0.30), followed by anger (≈ 0.19), surprise (≈ 0.16), disgust (≈ 0.15), fear (≈ 0.15), and joy (≈ 0.15)[cite: 3, 5]. The Afrikaans subset presents joy as the most frequent emotion (mean label rate ≈ 0.43), while surprise labels are entirely absent from its training split. For Igbo, anger is the most prevalent class (≈ 0.20), followed by disgust (≈ 0.19), sadness (≈ 0.17), joy (≈ 0.16), fear (≈ 0.08), and surprise (≈ 0.03). Neutral instances, derived from all-zero label rows, account for approximately 21.5% of the Igbo training set across all splits.

Two pretrained multilingual models are evaluated: **XLM-RoBERTa-large** and **AfroXLMR-large**. Both models are fine-tuned for five epochs using sigmoid outputs with binary cross-entropy loss for multilabel classification[cite: 6]. Performance is reported using macro, micro, and weighted F1 scores on the held-out test sets, with macro F1 serving as the primary evaluation metric owing to its sensitivity to underrepresented and minority emotion classes[cite: 6].

Four experimental conditions are systematically evaluated per language:

1. **Condition A – Baseline:** No augmentation; models are trained exclusively on the original human-annotated datasets (1,222 instances for Afrikaans; 2,945 instances for Hausa; 3,680 instances for Igbo)[cite: 6].
2. **Condition B – Back-translation:** Training data is expanded by adding high-resource pivot translations. For Afrikaans, this totals 3,030 instances via Dutch and English pivots. For Hausa, the training set is expanded to 4,417 instances by adding English (736 rows) and French (736 rows) back-translations[cite: 6]. For Igbo, English and French pivots via NLLB-200 produce 897 and 910 usable augmented rows respectively, expanding the training set to 5,487 instances.
3. **Condition C – Paraphrase-only:** Training sets are expanded via stochastic paraphrasing. Afrikaans is expanded to 3,033 instances. For Hausa, the corpus is expanded to 4,417 instances by adding 1,472 stochastically paraphrased rows[cite: 7]. For Igbo, paraphrasing via an Ollama-served LLM generates 1,836 additional rows, expanding the training set to 5,516 instances.
4. **Condition D – Combined:** The maximal data condition combining all augmentation paths. This expands Afrikaans to 4,041 instances, Hausa to 5,889 total training instances, and Igbo to 7,346 total training instances (Original + English BT + French BT + Paraphrasing)[cite: 7].

6.2 Results

Tables 1, 2, and 3 summarise test set performance across all conditions for Afrikaans, Hausa, and Igbo respectively. For Afrikaans, two runs produced anomalous near-zero scores due to suspected runtime label-alignment faults, specifically Condition B and C for AfroXLMR-large, and are omitted from analysis. For Hausa, all pipelines executed successfully across all conditions[cite: 6, 7]. For Igbo, all conditions completed successfully; however, the initial Condition D AfroXLMR-large run collapsed to near-zero F1 due to training instability under the heterogeneous 4x augmented corpus and required a full retraining run, which stabilised at a macro F1 of 0.4554 by epoch 5.

Cond.	Model	Mac	Mic	Wt
A – Base.	XLM-R	0.5996	0.6396	0.6492
A – Base.	AfroXLMR	0.6333	0.6550	0.6603
B – BT	XLM-R	0.5732	0.5977	0.6019
C – Para.	XLM-R	0.6768	0.7074	0.7134
D – Comb.	XLM-R	0.6531	0.6802	0.6862
D – Comb.	AfroXLMR	0.7302	0.7632	0.7661

Table 1: Test F1 scores (Afrikaans). Mac/Mic/Wt = macro/micro/weighted F1. XLM-R = XLM-RoBERTa-large; AfroXLMR = AfroXLMR-large. BT = back-translation; Para. = paraphrase. Conditions B & C (AfroXLMR) omitted due to a suspected runtime error.

Cond.	Model	Mac	Mic	Wt
A – Base.	XLM-R	0.5944	0.6109	0.6100
A – Base.	AfroXLMR	0.6793	0.6902	0.6896
B – BT	XLM-R	0.6403	0.6488	0.6492
B – BT	AfroXLMR	0.6003	0.6176	0.6140
C – Para.	XLM-R	0.6318	0.6418	0.6425
C – Para.	AfroXLMR	0.7125	0.7206	0.7198
D – Comb.	XLM-R	0.6458	0.6567	0.6545
D – Comb.	AfroXLMR	0.7010	0.7068	0.7077

Table 2: Test F1 scores (Hausa). Mac/Mic/Wt = macro/micro/weighted F1. XLM-R = XLM-RoBERTa-large; AfroXLMR = AfroXLMR-large[cite: 6]. Epoch validation limits are reported as proxy parameters where final test checkpoints are optimised[cite: 6, 7].

Baseline performance. Across all languages evaluated, both architectures establish highly competitive baselines. AfroXLMR-large consistently outclasses XLM-RoBERTa-large under baseline conditions (Condition A). In Afrikaans, AfroXLMR achieves a macro F1 of 0.6333 versus 0.5996 for XLM-RoBERTa-large. In Hausa, it reaches 0.6793 versus 0.5944[cite: 6]. In Igbo, AfroXLMR-large achieves 0.4858 versus 0.4500 for XLM-RoBERTa-large. The lower absolute scores for Igbo reflect the greater linguistic distance from high-resource languages in the pretraining corpora, as well as the prevalence of code-switching and informal social media text in the Igbo subset. This recurrent trend across all three languages confirms that AfroXLMR’s language-adapted pre-training provides a superior, linguistically grounded starting point for African morphosyntax and vocabulary representations.

Effect of back-translation. The downstream impact of back-translation reveals substantial architectural disparities. For XLM-RoBERTa-large, back-translation significantly lifts Hausa performance, jumping from a baseline macro F1 of 0.5944 to 0.6403 (+4.59 points)[cite: 6]. Conversely, it drops Afrikaans performance by 2.64 points (0.5996 → 0.5732). For AfroXLMR-large, back-translation causes severe performance degradation in Hausa, plunging its macro F1 score to 0.6003[cite: 6]. In

Cond.	Model	Mac	Mic	Wt
A – Base.	XLM-R	0.4500	0.5361	0.5168
A – Base.	AfroXLMR	0.4858	0.5663	0.5563
B – BT	XLM-R	0.4347	0.5204	0.4941
B – BT	AfroXLMR	0.4855	0.5621	0.5570
C – Para.	XLM-R	0.4841	0.5396	0.5404
C – Para.	AfroXLMR	0.5045	0.5841	0.5787
D – Comb.	XLM-R	0.4925	0.5584	0.5572
D – Comb.	AfroXLMR	0.4554	0.5288	0.5245

Table 3: Test F1 scores (Igbo). Mac/Mic/Wt = macro/micro/weighted F1. XLM-R = XLM-RoBERTa-large; AfroXLMR = AfroXLMR-large. BT = back-translation; Para. = paraphrase. Condition D (AfroXLMR) reflects a retrained run after an initial unstable convergence.

Igbo, back-translation yields mixed results: XLM-RoBERTa-large declines modestly from 0.4500 to 0.4347, while AfroXLMR-large is largely unaffected (0.4858 to 0.4855), suggesting that the Africa-adapted model is more robust to the noise introduced by English and French pivot translations. This suggests that routing text through intermediate high-resource pivot configurations can introduce detrimental semantic noise and label drift, particularly for general-purpose multilingual models.

Effect of paraphrasing. Paraphrasing proves to be a universally powerful and highly stable augmentation strategy. For XLM-RoBERTa-large in Afrikaans, it produces a +7.72 point improvement over baseline (0.5996 $\rightarrow$ 0.6768). For Hausa, the paraphrase-only condition (Condition C) combined with AfroXLMR-large unlocks the single highest primary metric score recorded for that language, peaking at a macro F1 of **0.7125**, an increase of 3.32 points over its baseline[cite: 7]. For Igbo, Condition C with AfroXLMR-large achieves the best result across all Igbo conditions at a macro F1 of **0.5045**, a gain of +1.87 points over the AfroXLMR baseline, and XLM-RoBERTa-large also improves substantially from 0.4500 to 0.4841. This consistent pattern across all three languages indicates that generating localised surface-level variety while keeping semantic structures tightly controlled provides a cleaner, highly dependable training signal.

Effect of combined augmentation. The combined multi-source augmentation condition (Condition D) reveals cross-lingual differences. For Afrikaans, combining strategies yields the absolute best overall configuration when paired with AfroXLMR-large, achieving a peak macro F1 of 0.7302 (+9.69 points over baseline). For Hausa, stacking all augmentations together expands the training set to 5,889 samples but slightly dilutes

peak performance: AfroXLMR-large reaches a macro F1 of 0.7010 under Condition D[cite: 7], which, while a major lift over the baseline, falls short of the isolated paraphrase-only run[cite: 6, 7]. In Igbo, the pattern is more pronounced: XLM-RoBERTa-large recovers to 0.4925 under Condition D, its second-best result, but AfroXLMR-large drops to 0.4554, well below its baseline. Inspection of training curves revealed that AfroXLMR-large converged to near-zero F1 in an initial unstable run and required retraining; after retraining it stabilised but was unable to match its paraphrase-only performance. Across all three languages, this behaviour demonstrates that combining heterogeneous augmentation sources can introduce optimisation instability, particularly for language-adapted models already sensitive to data distribution shifts.

6.3 Igbo: Qualitative Error Analysis

Error analysis was conducted on the best-performing Igbo configuration (AfroXLMR-large, Condition C). Three principal error categories emerged, corresponding to the linguistic dimensions outlined in Section 5.6.

Neutral class confusion. The neutral class produced the highest error counts in both directions: 388 false positives and 400 false negatives, yielding an F1 of only 0.36 despite being the second most frequent class. Because neutral is derived implicitly from the absence of all other emotion labels rather than through direct annotation, the model receives no explicit training signal distinguishing genuinely neutral content from ambiguous or under-expressed emotion. It therefore defaults to predicting neutral under uncertainty, and conversely misses neutral instances when surface-level emotional vocabulary is present. This finding is consistent with the Emotion Overlap and Ambiguity dimension outlined in Section 5.6.

Code-switching and social media noise. Igbo social media text frequently mixes Igbo with English and Nigerian Pidgin. Contrary to expectations, code-switched texts exhibited a lower error rate (44.9%) than pure Igbo texts (50.0%). A likely explanation is that English emotion words embedded in code-switched text provide direct lexical cues that the model’s multilingual representations exploit more readily than purely Igbo emotional expressions. Texts containing URL placeholders (##url##) showed elevated error rates (55.7%), consistent with the Linguistic Complexity dimen-

sion described in Section 5.6, as the placeholder occupies token budget without contributing semantic content. The mean text length of only 23.5 tokens leaves limited context for ambiguous or understated emotion expressions.

Class imbalance and multilabel difficulty. Instances carrying two or more simultaneous emotion labels produced an error rate of 96.6%, compared to 47.7% for single-label instances. With only 116 two-emotion instances in the test set, the model lacks sufficient training evidence to learn co-occurrence patterns reliably. The most prevalent substitution pattern was *joy* being collapsed into neutral in 63.8% of missed joy instances, suggesting that understated joy expressions in Igbo lack overt positive markers that the model can detect. This confirms the Class Imbalance Distortions concern raised in Section 5.6: even with paraphrase augmentation, minority co-occurrence patterns remain critically underrepresented.

7 Reflections

This multi-language comparative project revealed several critical insights about applying synthetic data generation to multilabel emotion classification pipelines across distinct African language families.

Paraphrasing consistently outclasses back-translation in lexical integrity. Across all three languages, paraphrasing strategies demonstrated vastly superior architectural stability compared to back-translation pipelines. Back-translation underperformed expectations and actively diminished downstream test accuracy for AfroXLMR-large in Hausa and XLM-RoBERTa-large in both Afrikaans and Igbo[cite: 6]. This recurring failure reveals that shifting low-resource text through distinct intermediate pivot typologies (e.g., Dutch, French, English) systematically introduces destructive semantic drift[cite: 6]. Nuanced emotional cues become generalised or translated incorrectly, corrupting the multilabel boundaries. Paraphrasing avoids this vulnerability by maintaining a single translation loop focused entirely on local surface vocabulary variety.

The interaction between model architecture and augmentation topology is non-linear. XLM-RoBERTa-large and AfroXLMR-large reacted differently to the expanded data environments depending on the target language. In Afrikaans,

AfroXLMR-large required the large, highly diverse 4x combined corpus (Condition D) to reach its optimal macro F1 score of 0.7302. In Hausa, AfroXLMR-large achieved its maximum performance using the isolated 2x paraphrase dataset (Condition C, 0.7125 macro F1) and experienced slight performance dilution when back-translated rows were introduced[cite: 6, 7]. In Igbo, both models responded best to paraphrasing (Condition C), while the combined condition (Condition D) caused a notable drop in AfroXLMR-large performance due to training instability introduced by the heterogeneous augmentation pool. This proves that models with language-adapted pre-training behave differently across typologically diverse targets, and developers cannot assume that larger data sizes automatically yield better scores.

Class imbalance and representation gaps remain persistent bottlenecks. Severe label skew was observed across all datasets: Afrikaans was heavily dominated by joy, the Hausa partition was significantly weighted toward sadness and anger[cite: 3], and Igbo exhibited a near-zero prevalence of surprise (2.6%), which was excluded from the label space entirely. Macro F1 calculations explicitly highlight this problem because they score all target emotions with equal weight[cite: 6]. Even though synthetic data expansions increased the absolute count of underrepresented emotions, models still struggled to differentiate boundary intersections for severe minority emotions. In Igbo, the derived neutral class alone contributed over 400 false negatives and 388 false positives in the best-performing configuration, underlining that implicit class derivation introduces additional ambiguity beyond that of annotated minority labels. This shows that uniform dataset scaling is helpful but ultimately insufficient unless combined with focused minority-class oversampling.

Pipeline validation and training instability require careful maintenance. Training instability was observed across multiple configurations. For Afrikaans, AfroXLMR-large failed entirely under Conditions B and C due to suspected label-alignment faults introduced during the NLLB-200 synthetic translation phase. For Igbo, AfroXLMR-large under Condition D required a full retraining run after an initial convergence to near-zero F1, ultimately recovering to 0.4554. Because multilabel text rows contain multiple binary flags, any parsing error or distribution shift introduced dur-

ing augmentation can decouple tokens from their ground-truth indicators. The fact that the Hausa data pipeline processed flawlessly across all configurations without runtime failure proves that strict pre-tokenisation validation, duplicate cleanup, and array checking are mandatory for reliable cross-lingual execution.

Synthetic data represents a bridge, not a total solution.

While augmentation generated meaningful performance gains across all three languages, yielding improvements of up to 13 macro F1 points for Afrikaans, peak scores above 0.71 for Hausa, and a +1.87 point gain for Igbo under the paraphrase condition, performance on heavily restricted emotions remained capped[cite: 7]. In Igbo, the multilabel error rate of 96.6% for two-emotion instances and persistent neutral class confusion demonstrate that augmentation addresses volume scarcity but cannot resolve annotation ambiguity or co-occurrence sparsity. This confirms that data augmentation is highly effective for low-resource environments but cannot entirely replace high-quality human annotations. Future frameworks should combine these data augmentation methods with cross-lingual few-shot prompting and emotion-stratified oversampling to push past these boundaries.

8 Conclusion

This project investigated whether data augmentation techniques—specifically multi-pivot back-translation and stochastic paraphrasing—can improve multilabel emotion classification performance for the low-resource African languages Afrikaans, Hausa, and Igbo using the BRIGHTER benchmark framework[cite: 1]. Two pretrained multi-lingual models, XLM-RoBERTa-large and AfroXLMR-large, were fine-tuned across four distinct experimental data boundaries[cite: 6].

Our empirical findings confirm that data augmentation provides meaningful performance improvements, but its ultimate success depends on matching the specific augmentation type with the right model architecture and target language. Paraphrasing proved to be the most consistent and reliable technique overall. It yielded a 7.72 point improvement for XLM-RoBERTa-large in Afrikaans, achieved the single highest test performance for Hausa when combined with AfroXLMR-large (macro F1 of **0.7125**)[cite: 7], and produced the best Igbo result of **0.5045** under the same con-

figuration. Back-translation showed high variance across all three languages, actively harming XLM-RoBERTa-large in Afrikaans and Igbo, and AfroXLMR-large in Hausa, due to semantic translation drift[cite: 6]. The maximum data condition (Condition D) achieved the overall best result for Afrikaans (0.7302 macro F1), while in both Hausa and Igbo it underperformed the paraphrase-only condition, demonstrating that scaling augmentation volume beyond a critical threshold can introduce noise that outweighs the benefits of additional data diversity.

These comparative discoveries provide actionable insights for developing robust, responsible NLP systems tailored to underrepresented African languages. They show that simple, targeted augmentation pipelines can successfully mitigate data scarcity, while underscoring that severe class imbalances and multi-label label integrity require careful handling. Future work should focus on designing emotion-stratified generation pipelines for minority labels, exploring cross-lingual transfer from related language families, and establishing automated validation frameworks to preserve multi-label text integrity during generation loops.

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